Module 2 – Mernstack - HTML

HTML Basics :

Theory Assignment

Question 1: Define HTML. What is the purpose of HTML in web development?

ANSWER : -

HTML (HyperText Markup Language) is the standard markup language used to create and structure content on web pages. It uses elements (tags) to organize text, images, links, multimedia, and other content so that browsers can display them properly.

Purpose of HTML in Web Development:

1. **Structures web content** (headings, paragraphs, lists, tables, forms).
2. **Creates hyperlinks** to connect web pages.
3. **Embeds media** like images, audio, and video.
4. **Acts as the foundation** for CSS (styling) and JavaScript (interactivity).
5. **Provides semantic meaning** to content for accessibility and SEO.

Question 2: Explain the basic structure of an HTML document. Identify the mandatory tags and their purposes.

ANSWER : -

A basic HTML document follows a standard structure:

```
<!DOCTYPE html>
<html>
<head>
   <title>Page Title</title>
</head>
<body>
   <h1>Hello World</h1>
   <p>This is a paragraph.</p>
</body>
</html>
```

Mandatory Tags and Their Purposes:

1. **<!DOCTYPE html>**
   - Declares the document type.

    - Tells the browser this is an HTML5 document.
2. **<html>**
    - Root element of the webpage.
    - Wraps all HTML content.
3. **<head>**
    - Contains meta-information about the page.
    - Includes title, charset, styles, and scripts.
4. **<title>**
    - Sets the webpage title (shown in browser tab).
5. **<body>**
    - Contains all visible content (text, images, links, etc.).

These tags form the basic framework of every HTML page.

Question 3: What is the difference between block-level elements and inline elements in HTML? Provide examples of each.

ANSWER : -

Block-Level Elements

- Start on a new line.
- Take up the full width available.
- Can contain other block and inline elements.

Examples:

-
- <p>
- <h1> to <h6>
- <section>
- <article>

Example:

```
<p>This is a paragraph.</p>
<div>This is a div block.</div>
```

Inline Elements

- Do not start on a new line.
- Take up only as much width as necessary.
- Usually used inside block elements.

Examples:

-
- <a>
-
-
- <img>

Example:

```
<p>This is <strong>important</strong> text.</p>
```

Key Difference:
Block elements create structure; inline elements format content within blocks.

Question 4: Discuss the role of semantic HTML. Why is it important for accessibility and SEO? Provide examples of semantic elements.

ANSWER : -

What is Semantic HTML?

Semantic HTML uses meaningful tags that describe the purpose of the content they contain.

Instead of:

```
<div class="header">></div>
```

We use:

```
<header></header>
```

Importance for Accessibility

- Screen readers understand semantic elements better.
- Helps visually impaired users navigate content.
- Improves overall user experience.

Example:

- <nav> clearly indicates navigation links.
- <main> identifies the main content area.

Importance for SEO (Search Engine Optimization)

- Search engines understand page structure better.
- Helps search engines index content correctly.
- Improves search ranking potential.

Examples of Semantic Elements:

- <header> – Top section of a page
- <footer> – Bottom section
- <article> – Independent content
- <section> – Thematic grouping
- <nav> – Navigation links
- <main> – Main content
- <aside> – Sidebar content

Lab Assignment

Task: Create a simple HTML webpage that includes:

- A header (), footer (), main section (), and aside section ().
- A paragraph with some basic text.
- A list (both ordered and unordered).
- A link that opens in a new tab.

ANSWER : -

```
<!DOCTYPE html>
<html>
<head>
  <title>Simple HTML Webpage</title>
</head>
<body>

  <header>
    <h1>Welcome to My Simple Webpage</h1>
  </header>

  <main>
    <h2>Main Content</h2>
    <p>
      This is a simple HTML webpage example demonstrating semantic elements,
      lists, and a link that opens in a new tab.
    </p>
```

```html
    <h3>Unordered List</h3>
    <ul>
      <li>HTML</li>
      <li>CSS</li>
      <li>JavaScript</li>
    </ul>

    <h3>Ordered List</h3>
    <ol>
      <li>Learn HTML</li>
      <li>Practice Coding</li>
      <li>Build Projects</li>
    </ol>

    <p>
      Visit
      <a href="https://www.example.com" target="_blank">
        Example Website
      </a>
      for more information.
    </p>
  </main>

  <aside>
    <h3>Side Information</h3>
    <p>This is additional information placed in the aside section.</p>
  </aside>

  <footer>
    <p>&copy; 2026 My Simple Webpage. All rights reserved.</p>
  </footer>

</body>
</html>
```

HTML Forms

Theory Assignment

Question 1: What are HTML forms used for? Describe the purpose of the input, textarea, select, and buttonelements

ANSWER : -

HTML forms are used to collect user input and send that data to a server for processing. They are commonly used for:

- Login and registration forms
- Contact forms
- Search boxes
- Surveys
- Online orders

Forms are created using the <form> element and contain various input controls.

Purpose of Common Form Elements

<input>

- Used to collect short user input.
- Has many types such as:
  - text
  - password
  - email
  - radio
  - checkbox
  - submit

Example:

```
<input type="text" name="username">
```

<textarea>

- Used for multi-line text input.
- Ideal for messages, comments, or descriptions.

Example:

```
<textarea name="message"></textarea>
```

<select>

- Creates a dropdown list.
- Contains <option> elements.

Example:

```
<select name="country">
  <option>USA</option>
```

```
  <option>Canada</option>
</select>
```

<button>

- Creates a clickable button.
- Can submit a form or perform actions.

Example:

```
<button type="submit">Submit</button>
```

Together, these elements allow users to enter and submit data.

Question 2: Explain the difference between the GETand POSTmethods in form submission.

ANSWER : -

When a form is submitted, data is sent to a server using either **GET** or **POST**.

GET Method

- Sends data in the **URL**.
- Data is visible in the address bar.
- Limited data size.
- Can be bookmarked.

Use GET when:

- Retrieving/searching data.
- Data is not sensitive.
- You want users to bookmark/share the URL.

POST Method

- Sends data in the **HTTP request body**.
- Data is not visible in the URL.
- Can send large amounts of data.
- More secure than GET (but still should use HTTPS).

Use POST when:

- Sending sensitive data (passwords, payment info).
- Submitting forms that modify data (create/update records).
- Sending large amounts of data.

Key Differences

Feature	GET	POST
Data Location	URL	Request Body
Visibility	Visible	Hidden
Data Size	Limited	Large
Security	Less secure	More secure

Question 3: What is the purpose of the labelelement in a form, and how does it improve accessibility?

ANSWER : -

The <label> element describes an input field.

Example:

```
<label for="email">Email:</label>
<input type="email" id="email" name="email">
```

Why <label> Is Important

Improves Accessibility

- Screen readers read the label when the input is focused.
- Helps visually impaired users understand what to enter.

Improves Usability

- Clicking the label focuses the associated input.
- Makes forms easier to use.

Better Form Structure

- Clearly connects text with form controls using the `for` attribute and matching `id`.

Summary

- HTML forms collect user data.
- <input>, <textarea>, <select>, and <button> are key form elements.
- GET retrieves data; POST submits sensitive or large data.
- <label> improves accessibility, usability, and clarity.

Lab Assignment

Task: Create a contact form with the following fields:

- Full name (text input) • Email (email input)
- Phone number (tel input) • Subject (dropdown menu)
- Message (textarea)
- Submit button

ANSWER : -

```
<!DOCTYPE html>
<html>
<head>
    <title>Contact Form</title>
</head>
<body>

    <h2>Contact Us</h2>

    <form action="#" method="POST">

        <!-- Full Name -->
        <label for="fullname">Full Name:</label><br>
        <input type="text" id="fullname" name="fullname" required pattern="[A-
Za-z\s]+" placeholder="Enter your full name">
        <br><br>

        <!-- Email -->
        <label for="email">Email:</label><br>
        <input type="email" id="email" name="email" required maxlength="100"
placeholder="Enter your email">
        <br><br>

        <!-- Phone Number -->
        <label for="phone">Phone Number:</label><br>
```

```
        <input type="tel" id="phone" name="phone" required pattern="[0-9]{10}"
placeholder="Enter 10-digit phone number">
        <br><br>

        <!-- Subject Dropdown -->
        <label for="subject">Subject:</label><br>
        <select id="subject" name="subject" required>
            <option value="">-- Select Subject --</option>
            <option value="general">General Inquiry</option>
            <option value="support">Support</option>
            <option value="feedback">Feedback</option>
        </select>
        <br><br>

        <!-- Message -->
        <label for="message">Message:</label><br>
        <textarea id="message" name="message" required minlength="10"
maxlength="500"
            rows="5"
            cols="30"
            placeholder="Write your message here..."></textarea>
        <br><br>

        <!-- Submit Button -->
        <button type="submit">Submit</button>

    </form>

</body>
</html>
```

HTML Tables

Theory Assignment

Question 1: Explain the structure of an HTML table and the purpose of each of the followingelements: <table>, <tr>, <th>, <td>, and <thead>.

ANSWER : -

```
<table>
    <thead>
        <tr>
            <th>Name</th>
```

```
            <th>Age</th>
        </tr>
    </thead>
    <tr>
        <td>Rohan</td>
        <td>22</td>
    </tr>
</table>
```

Purpose of each element

Element	Purpose
<table>	Defines the table
<tr>	Table row
<th>	Table header cell (bold, centered)
<td>	Table data cell
<thead>	Groups header content of table

➢ <**thead**> helps browsers and screen readers understand the header section.

➢ You can also have <**tbody**> and <**tfoot**> for better structure.

Question 2: What is the difference between col-span and row-span in tables? Provide examples.

ANSWER : -

Attribute	Direction	Function
colspan	Horizontal	Merge columns
rowspan	Vertical	Merge rows

Example of col-span:

```
<tr>
    <th colspan="2">Student Information</th>
</tr>
```

Example of row-span:

```
<tr>
    <td rowspan="2">Rohan</td>
    <td>Math</td>
</tr>
<tr>
    <td>Science</td>
</tr>
```

Question 3: Why should tables be used sparingly for layout purposes? What is a better alternative?

ANSWER : -

Tables were once used for webpage layout, but this is now discouraged.

<u>Problems with Using Tables for Layout:</u>

- Makes code complex and difficult to maintain
- Poor responsiveness on mobile devices
- Accessibility issues for screen readers
- Slower and less flexible design

Tables should only be used for **tabular data**, not page structure.

<u>Better Alternative for Layout</u>

CSS (Cascading Style Sheets)

Modern layout techniques:

- Flexbox
- CSS Grid

<u>These methods are:</u>

- Responsive
- Clean
- Accessible
- SEO-friendly

Lab Assignment

Task: Create a product Catalog table that includes the following columns:

- Product Name
- Product Image (use placeholder image URLs)
- Price
- Description

ANSWER : -

```
<!DOCTYPE html>
<html>
<head>
    <title>Product Catalog</title>
</head>
<body>

<h2 style="text-align:center;">Product Catalog</h2>

<table style="width:100%; border-collapse:collapse; text-align:center;"
border="1">

    <!-- Table Header -->
    <thead style="background-color:#f2f2f2;">
        <tr>
            <th colspan="5" style="padding:10px; font-size:18px;">
                Our Available Products
            </th>
        </tr>
        <tr>
            <th style="padding:8px;">Product Name</th>
            <th style="padding:8px;">Product Image</th>
            <th style="padding:8px;">Price</th>
            <th style="padding:8px;">Description</th>
            <th style="padding:8px;">Availability</th>
        </tr>
    </thead>

    <!-- Product Row 1 -->
    <tr>
        <td style="padding:10px;">Wireless Headphones</td>
        <td style="padding:10px;">
            <img width="40px"
src="https://avstore.in/cdn/shop/files/1.AVStore-Sonos-Ace-Front-Angled-View-
Hero-Black.jpg?v=1725620870"
                 alt="Wireless Headphones">
        </td>
        <td style="padding:10px;">$50</td>
        <td style="padding:10px;">
```

```html
            High-quality wireless headphones with noise cancellation.
        </td>
        <td style="padding:10px; color:green;">In Stock</td>
    </tr>

    <!-- Product Row 2 -->
    <tr>
        <td style="padding:10px;">Smart Watch</td>
        <td style="padding:10px;">
            <img width="40px"
src="https://rukminim2.flixcart.com/image/480/640/xif0q/shopsy-
smartwatch/j/j/u/1-99-android-ios-t800-ultra-smart-watch-1-9-48-mm-display-
original-imahycxv4apbzrut.jpeg?q=90"
                alt="Smart Watch">
        </td>
        <td style="padding:10px;">$120</td>
        <td style="padding:10px;">
            Feature-rich smartwatch with health tracking functions.
        </td>
        <td style="padding:10px; color:red;">Out of Stock</td>
    </tr>

    <!-- Product Row 3 -->
    <tr>
        <td style="padding:10px;">Bluetooth Speaker</td>
        <td style="padding:10px;">
            <img width="40px" src="https://encrypted-
tbn0.gstatic.com/images?q=tbn:ANd9GcR9cYWSaBNIhOPF2gguIun1olGtIYrHAlOO7w&s"
                alt="Bluetooth Speaker">
        </td>
        <td style="padding:10px;">$35</td>
        <td style="padding:10px;">
            Portable speaker with powerful bass and long battery life.
        </td>
        <td style="padding:10px; color:green;">In Stock</td>
    </tr>

</table>

</body>
</html>
```